3. Scenario: You are working on a project that involves analyzing a dataset containing information about houses in a neighborhood. The dataset is stored in a CSV file, and you have imported it into a NumPy array named house_data. Each row of the array represents a house, and the columns contain various features such as the number of bedrooms, square footage, and sale price.

Question: Using NumPy arrays and operations, how would you find the average sale price of houses with more than four bedrooms in the neighborhood?

Program:

```
import numpy as np
n = int(input("Enter number of houses: "))
house_data = []
for i in range(n):
    data = list(map(float, input("Enter bedrooms, square footage, sale price: ").split()))
    house_data.append(data)
house_data = np.array(house_data)
selected_houses = house_data[house_data[:, 0] > 4]
if len(selected_houses) > 0:
    average_price = np.mean(selected_houses[:, 2])
    print("Average sale price:", average_price)
else:
    print("No houses have more than 4 bedrooms")
```

Input:

```
Enter number of houses: 5
Enter bedrooms, square footage, sale price: 3 1500 220000
Enter bedrooms, square footage, sale price: 4 2000 350000
Enter bedrooms, square footage, sale price: 4 2200 260000
Enter bedrooms, square footage, sale price: 6 4000 505000
Enter bedrooms, square footage, sale price: 5 3500 290000
```

Output:

```
Enter number of houses: 5
Enter bedrooms, square footage, sale price: 3 1500 220000
Enter bedrooms, square footage, sale price: 4 2000 350000
Enter bedrooms, square footage, sale price: 4 2200 260000
Enter bedrooms, square footage, sale price: 6 4000 505000
Enter bedrooms, square footage, sale price: 5 3500 290000
Average sale price: 397500.0

...Program finished with exit code 0
Press ENTER to exit console.
```